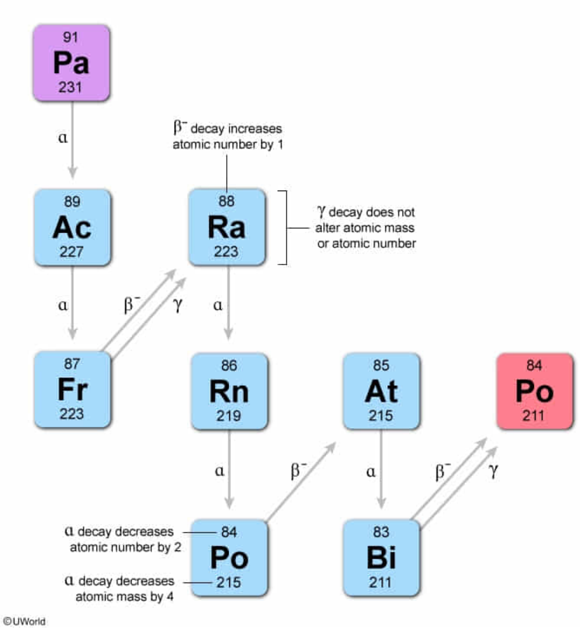
A radioactive atom decays by 5 alpha, 3 beta minus, and 2 gamma emissions to yield ^{211}Po. What was the original nucleus?

- A. ^{231}Pa [57%]
- B. ^{223}At [11%]
- C. ^{231}U [15%]
- D. ^{191}Ir [15%]

Correct
56% answered correctly

Explanation:



Radioactive decay is the process by which an **unstable nucleus becomes stable**. There are three common modes of decay, which change the nucleus in different ways:

- Alpha emission is the **release of a helium nucleus** (two protons and two neutrons). Each alpha emission reduces the atomic mass and atomic number of the atom by 4 and 2, respectively.

- Beta minus emission is the release of an electron from the nucleus and **turns a neutron into a proton**; it increases the atomic number by 1. The mass of an electron is negligible compared to the nucleus, so beta emission does not alter the atomic mass.

- Gamma emission is the **release** of a **high energy photon** that occurs when the protons and neutrons in a nucleus change configuration. It does not alter the atomic mass or atomic number of an atom.

The question states that ^{211}Po was generated after 5 alpha, 3 beta minus, and 2 gamma emissions. The 5 alpha emissions reduced the atomic mass by a total of 20, so the initial atom had an atomic mass of 231. Also, the original nucleus lost 10 protons from alpha emissions but gained 3 back from beta minus emissions, so it had 7 more protons than polonium. The gamma emissions had no effect on atomic mass or number. The periodic table shows that polonium has 84 protons, so the original nucleus must have had 91 protons. Therefore, the original nucleus was ^{231}Pa.

(Choice B) ^{223}At would be the original nucleus if the numbers of alpha and beta minus emissions were reversed.

(Choice C) ^{231}U would be the original nucleus if the numbers of beta minus and gamma emissions were reversed.

(Choice D) ^{191}Ir would be the product if ^{211}Po underwent the described decay series.

Educational objective:
Radioactive decay stabilizes atomic nuclei by releasing energetic particles. Alpha emission releases a helium nucleus (two protons and two neutrons), decreasing the atomic weight by 4 and the atomic number by 2. Beta minus emission releases a nuclear electron during the conversion of a neutron into a proton, increasing the atomic number by 1. Gamma emission is the release of a photon and has no effect on atomic mass or atomic number.

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Atomic Theory and Chemical Composition